



SCHOOL OF COMPUTER SCIENCE
Bachelor of Computer Science (Hons)
Bachelor of Software Engineering (Hons)

ITS66504: Systems Fundamentals

ASSIGNMENT 2

COVER PAGE

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Instructions to student:

- This is a **GROUP** assignment.
- Answers **MUST** be in PDF format.
- Submission due date is on the **28 DECEMBER 2024 11.59PM** via myTImeS.

Student declaration:

I declare that:

- *I understand what is meant by plagiarism*
- *The implication of plagiarism has been explained to me by our lecturer*
- *This assignment is our own work*

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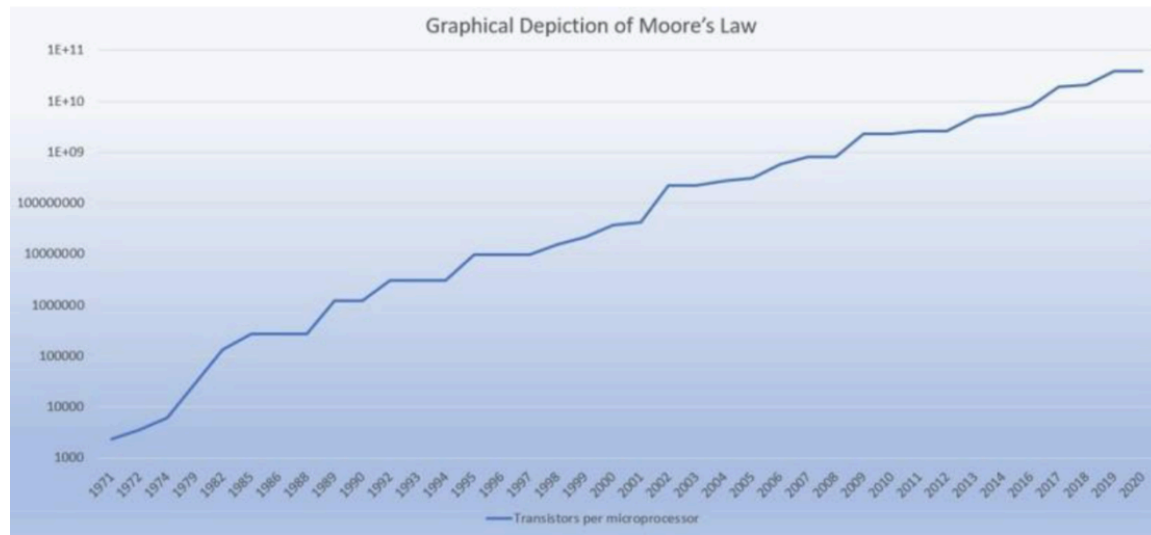
Question 1

For virtualisation

Virtualization is a technology that takes physical devices like operating systems, servers, storage, etc. and turns them into multiple virtual versions for use. With specific software, a computer's hardware can be partitioned into multiple independently running virtual machines. In my practice, using virtualization technology has helped us reduce hardware costs while increasing operational efficiency and Virtualization technology maximizes the use of physical hardware resources, allowing businesses to get a higher return on their hardware investment. Today, virtualization is a key pillar behind the economics of cloud computing. Surveys have shown that most organizations have significantly improved their network agility after using network virtualization. (Admin, 2023) Network virtualization enables cloud service providers to flexibly deliver scalable network solutions based on customer needs. This not only improves network security, but also effectively reduces operational costs. Network virtualization technology also plays an indispensable role in the IoT and enterprise networking space.

Moore's Law

Moore's Law points out how the number of transistors on a chip keeps growing rapidly. Moore predicted that this number would double roughly every two years, and this idea has shaped the semiconductor industry in a big way. Over time, it's helped bring down the cost of transistors while improving computing power and efficiency. Thanks to this steady progress, we've seen huge leaps in processing power and countless technological advancements.



Graph shows increase of transistors over the years in a microprocessor

Figure 1 (GeeksforGeeks, 2023)

Moore's Law has led to a substantial evolution in computing capability. Electronic device performance has increased exponentially as the number of transistors on a chip continues to rise. This expansion has been essential in facilitating sophisticated software programs, enhancing data processing power, and propelling technological advancement. Transistors' growing density enables computers to perform more tasks in less time, leading to quicker calculations and increased efficiency. The creation of increasingly sophisticated and potent software solutions has also been aided by this ongoing advancement.

Relationship

Moore's Law can be thought of as a “gas pedal” of technological development, revealing that the number of transistors on a chip doubles every two years. This phenomenon has fueled rapid growth in computing power while reducing the cost of producing technology. This has led to exponential growth in computing power and reduced costs. This rapid increase in hardware power has had a profound effect on the development and efficiency of virtualization technologies. Virtualization, on the other hand, is highly dependent on the performance of the underlying hardware. It enables multiple virtual machines to run on a single physical device by emulating hardware resources through software. Each VM has its own operating system,

although they share the same hardware resources. So Moore's Law has driven an increase in transistor density, making processors faster and more powerful, thus allowing modern hardware to handle multiple VMs at the same time without significant performance degradation. Hardware driven by Moore's Law such as multi-core processors further enhances virtualization capabilities by enabling parallel processing. In addition to relying on Moore's Law, virtualization prolongs its "life." Organizations can make better use of their current gear with virtualization, eliminating the need to constantly buy new hardware. Virtualization is widely used by cloud providers like Microsoft Azure and Amazon AWS. They can provide distinct resources for every user by running virtual computers for several users on a single server. Moore's Law is slowing down as transistor sizes get closer to their physical limits, even if the hardware performance improvements it brings are crucial for virtualization. As a result, the development of virtualization technology faces certain obstacles. Moore's Law and virtualization go hand in hand. Moore's Law keeps hardware performance improving, allowing virtualization technology to evolve, which in turn helps organizations use hardware resources more efficiently. For our students taking this course, understanding this relationship can help to better understand how modern computing technologies work.

Virtualization has three different types of virtualization: Full Virtualization, Para-Virtualization, and OS-Level Virtualization.

Full virtualization

Full virtualization involves consolidating physical computing resources into one or more virtual instances. Any software or operating system that is compatible with the underlying hardware can run within a virtual machine because each instance runs in its own virtual environment.

Hardware-assisted virtualization and binary translation are the two methods most frequently employed for full virtualization. To guarantee compliance with the virtual machine, binary translation dynamically changes some software instructions, substituting safe operations for risky ones. Conversely, hardware-assisted virtualization makes use of processor capabilities to enable the isolation of guest operating systems, frequently without necessitating major modifications to the guest OS. (Wikipedia contributors, 2024) Analyze architecturally there are advantages and

disadvantages about full virtualization, its advantages are significant: each virtual machine is completely isolated from each other so it ensures the security and fault tolerance of the system because it prevents errors and crashes in one virtual machine from affecting another virtual machine. And full virtualization has flexibility and compatibility, flexibility in the sense that it can support running unmodified operating systems of all kinds, and compatibility in the sense that it is suitable for running outdated or incompatible operating systems on modern hardware. But full virtualization has some drawbacks and challenges such as hardware emulation and instruction conversion introduces latency, implementation and management of full virtualization requires more specialized skills so it increases its complexity.

From a technical analysis, full virtualization relies on hardware and software support to enhance its benefits such as efficiency and compatibility. Modern cpus reduces the performance loss of instruction emulation by providing hardware-assisted virtualization. And in the absence of hardware support, the hypervisor needs to capture and translate the instructions issued by the guest operating system so that they can be executed smoothly on the host operating system. Full virtualization is also suitable for environments that require a high degree of isolation, such as in scenarios where highly sensitive data is handled, as each virtual machine is completely independent of the other, avoiding interferences between the different virtual machines and ensuring security and isolation, but this may lead to a reduction in performance. Fully virtualized representative products such as VMware and Oracle VirtualBox.

Para-Virtualization

Para-virtualization is a virtualization technique that requires a virtual machine's operating system to be modified to match the underlying virtualized environment to improve performance. This technique reduces the overhead of virtualization by avoiding full hardware emulation.

Architecturally speaking the advantage of Para-virtualization is that the operating system has been modified to support the virtualized environment so it can communicate directly with the underlying hypervisor, avoiding the overhead of hardware emulation and therefore improving performance, and Para-virtualization reduces the consumption of system resources by optimizing the interfaces resulting in better resource utilization. Finally, Para-virtualization supports

multiple virtual machines running on a single hardware platform and ensures efficient resource allocation. However, it has its drawbacks. Its drawback is that it requires modification of the operating system, and if it is found that the operating system does not support it, it cannot continue to be used, so because it requires modification of the operating system code, its isolation is naturally not high, and its compatibility is poorer than that of full virtualization. From a technical point of view it is not necessary to fully emulate a hardware device, but rather it is achieved through a direct link between the virtualization hypervisor and the operating system. So by the above judgment and analysis Para-virtualization is suitable for scenarios that require high performance but relatively low isolation requirements.

OS-Level Virtualization

The architecture of OS-Level Virtualization is characterized by the fact that it is virtualized by creating multiple isolated user-space instances on a single operating system kernel. Each container runs on the same operating system kernel but remains independent of each other. The advantage of OS-Level Virtualization is that the containers are directly on the host operating system, which avoids overhead and improves performance. It is also faster to boot than other virtualization operations and significantly reduces wasted resources. However, there are many limitations such as in OS-Level Virtualization all containers share the host operating kernel, which means that containers can only run the same operating system as the host, so its kernel is limited. Its isolation and security are also relatively weak compared to full virtualization and it cannot support different operating system environments. There are many application scenarios such as microservice architectures, DevOps and continuous delivery, and cloud computing platforms. So from a technical point of view it is suitable for use in situations where high performance and fast startup environments are required.

Question 2

Introduction

Medicare Solutions is one of the best medical diagnostic centers in Singapore dealing with radiology, pathology, and diagnostic imaging. In ten years, it has amassed a huge amount of data in terms of patients and their imaging records. Increasing patients and developed services have

put the center into renovating the infrastructure. Therefore, they are facing two challenges: Compatibility and Cybersecurity concerns.

MediCare Solutions are cautious about upgrading their workstations and imaging systems due to compatibility concerns regarding their outdated diagnostic software and historical patient data. Compatibility issues prevail when software and hardware cannot operate effectively with each other or the operating system, hence causing errors, crashes, performance issues, or security risks (LinkedIn, n.d.) - for *How do you solve software compatibility?*. Hence, the compatibility of programs, systems and devices are essential for ensuring optimal system performance and user experience.

This challenge can be addressed by using hypervisor technology as it allows hardware emulation. According to Alibaba Cloud (2024), hardware emulation refers to the process of creating a software model that mimics every behaviour and function of a hardware device, replicating its original functionality. This allows software designed for one system to run on a completely different system. Hence, hypervisor technology can emulate hardware for each virtual machine by creating virtual environments that replicate the behaviour and functions of the host system. This provides each virtual machine access to its own set of virtual hardware components, including virtual central processing unit (CPU), memory, storage, and other devices, which saves costs for additional hardware purchases (JBS.live, n.d.). In the case of MediCare Solutions, hypervisor can create virtual machines that imitate older hardware, allowing MediCare to run legacy software on the upgraded systems and access historical patient data without relying on outdated physical equipment. Therefore, MediCare Solutions can upgrade workstations and imaging systems to improve their performance, reliability, and service while maintaining compatibility with legacy software through hypervisor technology.

Another challenge that concerns MediCare Solutions in installing new software applications on their production systems is due to cybersecurity risks. According to Ribeiro (2024), the weekly average number of cyberattacks on each healthcare organization rose to 2,018 between January and September 2024, representing a 32% increase compared to the same period in the previous year. Moreover, data breach such as exposing sensitive patient data, and

healthcare-specific malware like ransomware are among the list of top 7 healthcare cybersecurity threats (Terranova Security, 2024). With the surge in cyber attacks in recent years, these statistics underscore the importance of cybersecurity for MediCare Solutions, justifying their caution in installing new software on production systems.

Hypervisor technology can be used to address the challenge. Hypervisor is a type of virtualisation software that creates, runs and manages multiple virtual machines (VMs), dynamically allocating resources such as central processing unit (CPU) and memory, to each virtual machine as required (Wind River, n.d.).

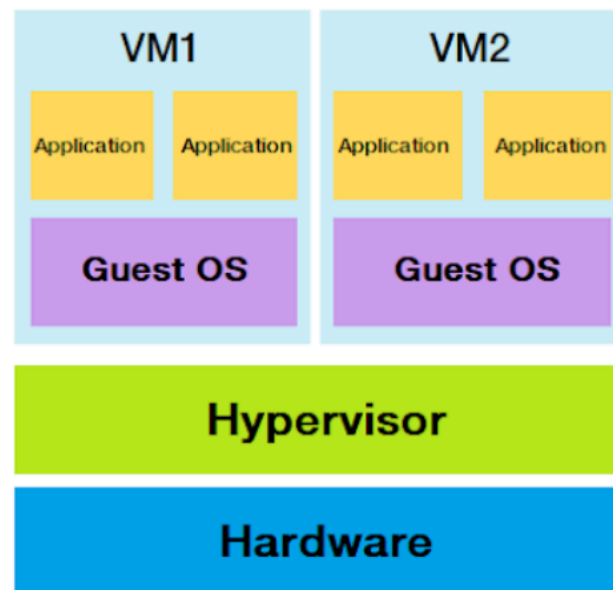


Figure 2: Type-1 Hypervisor (Network Urge, 2017)

Figure 2 illustrates the architecture of a hypervisor-based virtualized environment where the hardware forms the “base layer”, and the individual virtual machines are the “top layer”, running on top of the hypervisor, which is allocated between the hardware and the virtual machines. This enables the hypervisor to isolate the virtual machines from one another and the hardware system, allowing the virtual machines and the hardware to communicate (for resource allocation) exclusively through the hypervisor only (Susnjara & Smalley, 2024). In addition, each virtual machine provides an isolated environment consisting of an operating system and several applications based on Figure 2. This allows multiple virtual machines to run different operating systems and applications simultaneously on a single computer without affecting the rest of the host system. Isolation limits the access of virtual machines to certain resources, thus reducing the

risks of facing unforeseen cyber attacks such as unauthorized access, data breaches, or malware issues (LinkedIn, n.d.) - for *What are the pros and cons of isolating vs eradicating infected devices?*. Furthermore, the hypervisor can encrypt the data stored on the virtual machines to protect patient data, or use secure boot to verify the integrity of the virtual machines (LinkedIn, n.d.) - for *How does a hypervisor improve security isolation?*. Therefore, the isolation characteristic of hypervisor technology can prevent the loss of patient data, healthcare-specific malware and cyber attacks, which can address MediCare Solutions' concerns regarding cybersecurity risks when installing new software applications on their production systems.

Select a suitable hypervisor for MediCare Solutions

There are two types of hypervisors: Type-1 and Type-2. Each type of hypervisors offers different features that are ideal for specific applications and environments.

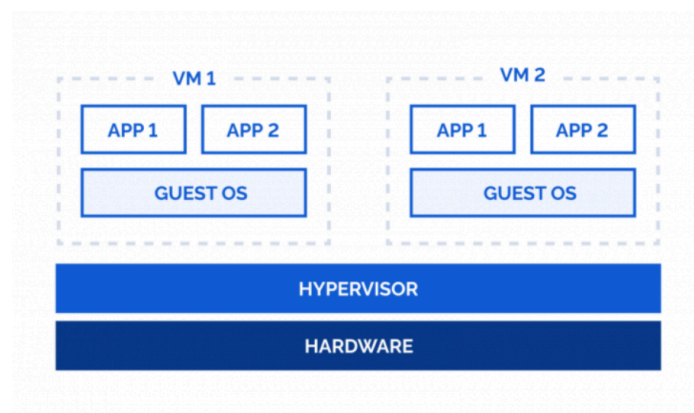


Figure 3: Type-1 Hypervisor (Pankevych, 2023)

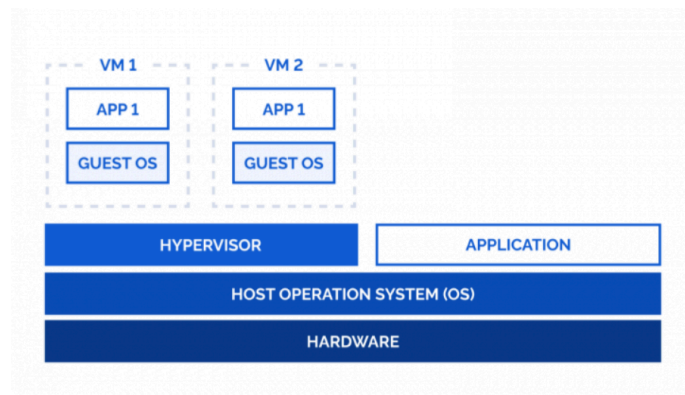


Figure 4: Type-2 Hypervisor (Pankevych, 2023)

According to Figure 3, Type-1 hypervisor, also known as a bare-metal hypervisor, operates directly on the computer's physical hardware, having direct access and more control over the computer hardware and its resources such as central processing unit (CPU), memory, and storage (SentinelOne, 2024). However, according to Figure 4, Type-2 hypervisor, also known as a hosted hypervisor, operates as an application on top of a host operating system, which is allocated between the hypervisor and the physical hardware. This means that the hosted hypervisor has to rely on the host operating system to access the physical hardware and its resources. Moreover, compared to bare-metal hypervisors, other applications can run alongside the hosted hypervisor on top of a host operating system which is independent of the virtual machines.

Considering MediCare Solutions' circumstances and concerns, a Type1 hypervisor is intended for them as it offers direct access to the hardware resources. Direct access to physical resources allows the implementation of custom resource allocation strategies to service each virtual machine, offering greater performance for virtual machines and their workloads (AWS, n.d.). Hence, unlike Type-2 hypervisors, Type-1 hypervisors provide maximal performance for large-scale workloads which reduces latency and optimizes resource allocation. MediCare Solutions is a medical diagnostic center that requires the documentation of vast amounts of patient diagnostic data and medical imaging records. This shows that the center involves large, resource-intensive workloads that require the use of Type-1 hypervisors for more reliable and consistent performance, reducing latency.

In addition, due to the need to modernize and upgrade MediCare Solutions' workstations and imaging systems, Type-1 hypervisors are scalable in handling increasing amounts of virtual machines and workloads. Conversely, Type-2 hypervisors run on top of a host operating system, causing limitations in managing hardware resources directly, hence resulting in reduced scalability when dealing with large-scale workloads. As MediCare Solutions expand its services, Type-1 hypervisors can handle a wide variety of virtualized environments with ease, allowing the center to expand their operations efficiently as compared to Type-2 hypervisors (Technologies, 2024).

In order to integrate legacy software with modern systems and access historical patient data, hardware emulation is needed, which may affect the performance of a Type-1 hypervisor because emulation will involve additional processing steps that lead to a reduced performance in utilizing the hardware capabilities directly (Patel, 2024). Although Type-1 hypervisors prioritize optimal performance and scalability due to its direct access to the physical hardware, hardware emulation can still be included when needed to ensure compatibility between guest operating systems or applications with the host system. Hence, even though there may be reduced performance because of hardware emulation, Type-1 hypervisors are still ideal for the increasing amounts of virtual machines and workloads, in managing growing demands, and upgrades on MediCare Solutions' workstations and imaging systems.

Lastly, Type-1 hypervisors offer enhanced isolation for each virtual environment. The reason is because there is no shared layer like the host operating system for Type-2 hypervisors, hence any vulnerability such as security malware, data breaches and unauthorized access that is present in the host operating system can directly impact all virtual machines negatively (Bigelow & Kirsch, 2024). Therefore, Type-1 hypervisors are significant in providing robust security for each virtual machine, protecting sensitive information and minimizing the risk of patient data loss, healthcare-specific malware and cyber attacks, which can address MediCare Solutions' concerns about cybersecurity risks when installing new software applications on their production systems.

In conclusion, Type-1 hypervisors can address the challenges and concerns (compatibility and cybersecurity) of MediCare Solutions. They are ideal for MediCare Solutions as they prioritize and ensure higher performance, scalability and security in managing large-scale workloads and virtual machines as compared to Type-2 hypervisors.

Question 3

Introduction

GlobalEd Solutions is an education technology company that focuses on interactive learning platforms for schools. Recently, the company was working with the development teams in North

America, Europe, Asia, and Africa. However, they encountered on the integration issues. The development teams were working in separate virtual machines or local environments. This is leading to an incompatibility issue when integrating tools and content into the platforms. Thus, the centralized team had to spend many times to reconfiguring and retesting and this will cause the product launch to be delayed.

In order to solve the integration issue, GlobalEd Solutions planned to build a centralized virtualized development and testing platform using the hypervisor technology. Through this platform, the centralized team will be able to provide a unified development environment for all development teams. In this way, the integration issue will be solved and speed up the product launch timeline.

Application of hypervisor technology

Hypervisor is a virtualization technology, it allows multiple virtual machines to run on a single physical host. Each virtual machine has a separate operating system and application environment. The technology not only improves resource utilization, it also provides isolation and manageability for each development team. Based on that, a centralized virtualization platform will unify the management of all development and testing environments. The local virtual machines or local environment of each development team will be consolidated into one virtualized system. As a result, the development team will be able to access and utilize preconfigured development and test resources through the platform and increase productivity.

There are two types of hypervisors which are Type-1 hypervisors and Type-2 hypervisors. Type-1 hypervisors are more suitable for the GlobalEd Solutions company. Practical examples of Type-1 hypervisors are VMware and Microsoft Hyper-V. These hypervisors run directly on the hardware and provide high performance and reliability. Practical examples of Type-2 hypervisors are Oracle VirtualBox and VMware Workstation, which run on the operating system and it is more suitable to the virtualization needs of small companies or individual users.

With Type-1 hypervisor, GlobalEd Solutions will use the high performance and security of it to meet the needs of organizations. The hypervisors will directly interact with the hardware,

reducing the operating system layer and optimizing the use of the resources. Besides that, the hypervisor also ensure isolation between virtual machines. Sensitive data of each development team will not be leaked out. This architecture also supports multiple virtual machines simultaneously. It allows each development team to develop and test multiple modules at any time, increasing efficiency of the work.

In addition, preconfigured VM images enable consistent tools that can be use by different development teams. As a result, the time that spent by the centralized team to configure conflicts and version incompatibilities will be reduced. The hypervisor snapshot and clone feature can increase efficiency. It also allows the development team to save the progress of their work anytime and quickly roll back errors. With centralized management, development teams will be able to allocate resources flexibly and increase productivity.

Mechanisms for speed up time to market

Hypervisor technology speeds up the product of GlobalEd Solutions through several mechanisms. One of the mechanisms is the dynamic resource allocation feature. It can dynamically adjust the CPU, memory, storage resources of the virtual machine based on the developer needs. This will ensure that each team can finish their work efficiency. It improves the overall development efficiency and optimizing resource utilization.

Next, the automation tool of the hypervisor is integrated with the CI/CD pipeline. It is able to address testing and deployment. This feature not only reduces the time that spent manually on errors, it also helps to increase the speed of version iteration.

Moreover, the hypervisor snapshot and clone feature provide effective support for speed up the product launch timeline. The clone feature allows development teams to quickly copy an existing virtual machine environment without rebuild a new one. Besides that, it also provides separate environment for the development to testing different modules. Each module can run in a separate virtual machine, it helps avoiding competition for resources between teams. This feature also allows multiple modules to work at the same time, it reduces dependency between modules and accelerates the time to market.

Last, centralized virtualization platforms also facilitate collaboration between distributed teams. All teams have real time access to the same virtualized environment. This helps reduce communication issues between teams and also reduce delays caused by their different locations.

Additional benefits of hypervisor technology

Hypervisor technology helps improve the utilization of hardware resources. Virtualization technology, which can run many virtual machines using a single physical host. It reduces hardware requirements and energy consumption, helps in lowering the cost of operations. Besides that, it also reduces the impact on the environment, achieving the goal of sustainable development.

Other than that, hypervisor technology enhances system flexibility and scalability. The cloning feature of hypervisor technology helps teams when adding a new development or test instances. The feature can quickly replicate existing virtual machine environments, without having any purchase on additional hardware. This flexibility feature supports multiple projects simultaneously, allowing teams to quickly adapt to changing project requirements.

In addition, hypervisor technology also improves the system reliability and fault tolerance. With the hypervisor snapshot feature, the development team can save their work progress anytime and quickly roll back to previous stable state when encountering errors. This feature will effectively reduce problems caused by unexpected errors and improve work efficiency.

Meanwhile, a centralized virtualization platform has high security. Virtual machines will be segregated from each other. This will ensure that security issues in one virtual machine will not spread to others virtual machines or the entire system. This is an important benefit for GlobalEd Solutions company which focuses on the educational technology platforms that may handle sensitive data. Besides that, with centralized management, development teams are able to use a unified security policy and reduce security issues caused by human intervention.

Last but not least, hypervisor technology also supports the legacy systems. Many educational institutions may still use the older versions of software or operating systems. Hypervisor

technology can run these types of legacy systems through virtual machines, allowing for the connectivity with modern technology. This compatibility increases the suitability of platforms.

Conclusion

GlobalEd Solutions use hypervisor technology to solve the problem of system integration. A unified development and test environment will speed up time to market and reduce the workload of the centralized teams. Snapshot and cloning features will help in simplifying the testing and rollback process, while the dynamic resources allocation and centralized management will improve the hardware resource utilization.

Besides that, CI/CD integration also will reduce the development time, enabling the GlobalEd Solutions company to quickly adapt to market demands and improve the product quality.

In the end, hypervisor technology also brings many additional benefits. These include reduced operating costs, increased system flexibility, fault tolerance, security and support legacy system integration. Through hypervisor technology, GlobalEd Solutions companies are able to solve the problems and achieve breakthroughs in operational efficiency and market competitiveness.

Question 4

Introduction

GreenBites Cafe, a famous store in Malaysia that has been open for a long time, was experiencing frequent outages and performance issues due to a heavily aging server network, and they wanted to make sure that both the physical store and the online platform were up and running. With limited funds and a small IT team, the café had to extend the lifespan of its current hardware, and GreenBites Cafe primarily wanted to reduce expenses, improve manageability, and centralize control of its offline and online operations. Their main challenge was to transition from the current to a virtualized network infrastructure with a manageable budget.

How a Hypervisor Helps with Network Virtualization

A hypervisor, particularly a Type-1 (bare-metal) or Type-2 (hosted) hypervisor, plays a crucial role in virtualizing the underlying hardware resources.

We can use virtualized network components, such as virtual switches and routers, to enable GreenBites Cafe to create a centralized management system. IT staff can monitor and manage all networked systems from a single interface, which reduces the complexity of managing multiple brick-and-mortar locations and online platforms.

The hypervisor can dynamically and rationally allocate various resources (such as CPU, memory and storage resources) among virtual machines, which ensures optimal utilization of available resources. This flexible allocation helps to maximize the lifespan of aging servers, thereby delaying the need to replace them.

The hypervisor enables multiple virtual machines (VMs) to operate on a single physical server so that GreenBites Cafe can allow multiple applications (such as ordering systems, inventory management systems and customer feedback systems) to be consolidated onto fewer physical servers. This greatly reduces hardware dependency and largely reduces the cost of maintaining the equipment.

Virtual machines can be migrated between physical servers in real time without downtime, ensuring business continuity even during hardware maintenance and failures

Centralized Control Over Applications and Desktops

With a virtualized desktop environment, GreenBites Cafe can deploy a virtual desktop infrastructure so IT staff can remotely manage updates, deploy new software and troubleshoot systems, reducing time costs and complexity.

Centralized control allows for better securing of data and execution commands, for example, enabling the protection of customers' sensitive data and operations. Virtualization can isolate applications and workloads, reducing the risk of system-wide vulnerabilities.

The hypervisor ensures consistent application and system performance through a standardized virtual environment, which allows GreenBites Cafe to run successfully in different environments and makes it easier and more convenient to manage and operate.

Maximizing IT Hardware Infrastructure Investment

The hypervisor does this by running lightweight virtual machines instead of running large operating systems and applications directly. Doing so allows older servers to be used less frequently and still report productivity. Even with aging hardware, GreenBites Cafe enables high performance and can defer high expense equipment upgrades.

Virtual machines are highly scalable, and as content and services increase, we don't have to buy new physical servers to add and create more new virtual machines to use. This scalability is perfect for managing peaks in demand such as large promotions, co-branding and holidays.

By consolidating work onto fewer physical servers, we can reduce the need for hardware repair and replacement.

We use fewer physical servers, which means we consume less energy and save on cooling costs, contributing to GreenBites Cafe's eco-friendly image.

The general benefits of Type-2 hypervisor to GreenBites Cafe company

A Type-2 hypervisor, also known as a managed hypervisor, runs on top of an existing operating system. Since GreenBites Cafe they have a limited budget and a small IT team and aging hardware, there are significant advantages to using a Type-2 hypervisor.

Because the Type-2 hypervisor runs on currently available operating systems, GreenBites Cafe can utilize existing servers and workstations without requiring extensive hardware updates. So no specialized hardware is required. And many Type-2 hypervisors are available for free or at a

low cost, making them more suitable for small businesses with limited budgets. It's more affordable.

By virtualizing work content, GreenBites Cafe can reduce the frequency of aging physical servers and can run multiple virtual environments on a single machine, reducing the number of physical servers. Virtualization allows multiple in-store applications to be consolidated on top of fewer physical devices.

Type-2 VM Manager requires no complex configuration or additional infrastructure and can be installed directly on existing computers. IT staff with only basic operating system knowledge can quickly deploy and run virtual machines, saving time and training costs.

The Type-2 hypervisor can support multiple operating systems on a single physical server, enabling GreenBites Cafe to run multiple applications. And virtual machines can be easily moved between devices, which can reduce a company's effort to migrate without downtime.

The Type-2 hypervisor is not specifically designed for applications that manage a large number of virtual machines or complex network infrastructures at the same time. It is more suited to smaller application scenarios, such as a small number of virtual machines or localized IT management needs. This means that GreenBites Cafe can use it to better control its IT systems, such as managing applications, desktop environments, or workloads in each store. For a smaller company with limited resources like GreenBites Cafe, this is a practical and efficient way to meet its current needs.

As the business grows or the financial situation improves, GreenBites Cafe can shift programs to implement a more robust system. For example, Type-1 hypervisor or cloud-based virtualization, while retaining the Type-2 hypervisor setup and infrastructure.

Question 5

Select a fuller-featured solution

My company manager chose a cheaper 'good enough' virtualisation solution based on economic costs, but we decided to persuade him to switch to a more comprehensive and higher-cost solution that is more in line with our company's actual needs based on our company's needs. We decided to persuade the manager from the perspective of long-term cost-effectiveness, company needs, employee experience and customer satisfaction, performance and efficiency.

In terms of finance, although choosing a more expensive virtualisation solution will lead to a higher initial investment, the subsequent long-term cost savings from reduced maintenance and the need to replace equipment less frequently will make up for the initial shortfall. If we just choose a 'good enough' solution, as the company expands, we may be forced to replace it due to technical limitations, resulting in additional investment costs. If we use a more comprehensive solution from the beginning, it can better support business expansion and bring higher returns to the company. In addition, the 'good enough' solution may cause safety risks due to insufficient performance, resulting in inefficiencies, data loss and other safety issues. The cost of maintenance may be much higher than the additional cost of using a comprehensive solution from the beginning. Therefore, from a financial perspective, choosing a more comprehensive virtualization solution can avoid many hidden costs and help the company better complete business expansion.

In terms of company needs, we have used a more comprehensive solution from the beginning. Although it has caused additional expenses for the company, in the current trend of social development, our company will face many changes in business growth and demand. Our company also has some plans for future expansion. A more comprehensive solution can bring higher scalability and business opportunities, with extremely high scalability, which can better meet the company's business needs. From the perspective of business competition, if our competitors use more comprehensive and advanced solutions and we do not, we will lose competitiveness, which will lead to the loss of customers and a decline in the company's

economic benefits. With a more comprehensive solution, the company will have a better brand image in the market and demonstrate the company's innovation and technical capabilities. These hidden capabilities will likely bring higher returns for the company and attract potential customers in the future. From these aspects, adopting a more comprehensive solution is more in line with our company's needs.

In terms of the adoption of solutions from the perspective of both employees and customers, in the case of a more comprehensive solution, our company's system will have higher automation technology, so that our employees can reduce the learning time when they come into contact with the company's business, while improving the efficiency of employees. At the same time, a more comprehensive solution often has higher security, which can prevent data loss and system crashes. If a 'good enough' solution is chosen, we may need technical staff to learn this technology for maintenance, while a more comprehensive solution can avoid this to a certain extent, thereby improving employee productivity. In another respect, more comprehensive solutions can provide smarter technical support when employees work in teams, simplify work processes, and greatly improve team efficiency and motivation. From these perspectives, companies that use advanced technology can better improve employee motivation, and a modern work environment can also attract highly skilled technicians, retain talent, and enhance the company's reputation. On this basis, a more comprehensive solution will bring stronger business capabilities, which reflects the company's innovation capabilities and brand image. We can better meet customer needs, thereby improving customer satisfaction and even bringing in more hidden customers.

Finally, we consider it from the perspective of virtualised performance. The core of virtualisation technology lies in the integration of resources. A more comprehensive solution can be managed more efficiently, reducing hardware waste. A 'good enough' solution may lack advanced resource allocation capabilities, resulting in underground hardware utilisation and resource waste. From the perspective of performance scalability, a more comprehensive virtualisation solution can support dynamic scaling capabilities. Adding resources such as virtual machines when the business grows, while a 'good enough' solution may require manual upgrades when faced with business growth, resulting in wasted time and additional costs. From the perspective of security, a more comprehensive solution will also have higher security functions, such as

emulator monitoring, virtual machine isolation and encryption, and other virtualization security strategies, while a ‘good enough’ solution may not be able to completely isolate resources between different virtual machines, thus posing a security risk. When it comes to server virtualization, a more comprehensive solution can support three types of virtualization and switch between them in a timely manner according to business needs, thus meeting a variety of complex business scenarios. A ‘good enough’ solution may not be able to support all types, resulting in wasted performance utilization and management efficiency, and it may not work properly when faced with complex scenarios. Finally, from the perspective of the hypervisor we can see that a more comprehensive solution can support Type-1 hypervisors, which have higher performance and lower latency, as well as higher security. A ‘good enough’ solution may only support Type-2 hypervisors, which may not be able to meet the requirements for performance and stability in a production environment. Our company has high requirements for enterprise production, and Type-2 is only suitable for lightweight enterprises. From these perspectives, choosing a more comprehensive solution is more in line with our company's needs.

By thinking from multiple perspectives, we found that a more comprehensive solution would be better suited to our company's needs and development.

Select ‘good enough’ solution

The business manager had considered a more comprehensive virtualisation solution for performance, but our company is just a small start-up. We decided to persuade the manager to choose a ‘good enough’ solution, taking into account cost-effectiveness, company needs, employee capabilities, and virtualisation technology.

First, from a cost-benefit perspective, as a small new enterprise, a comprehensive virtualization solution, although it has better performance, also comes with many expensive investments, such as software license fees, high-performance hardware, etc., which are a high burden for the enterprise. Even if the enterprise needs to finance or pay in installments to spread the initial cost, and then once the business encounters subsequent operational errors, all of this will become a burden on the business. A ‘good enough’ solution has a lower initial investment and only needs

to meet the current needs. The business may have more sufficient funds for market development and advertising and other business.

From the perspective of company needs, as a small enterprise, we don't need too many additional functions. Although comprehensive solutions have some stronger performance and more comprehensive functions, for enterprises, many functions are not needed in the short term, and 'good enough' solutions, although they do not have comprehensive functions at the beginning, can gradually add functions as the company's business needs grow, thus avoiding waste of resources. At the same time, as the company's business expands, the 'good enough' solution can be upgraded as the business expands, which can provide greater flexibility and choice, avoiding the high cost of a one-off investment and the risk of future business adjustments. At the same time, as technology is constantly being updated and replaced, initially selecting only a basic solution allows us to use more advanced technologies and solutions in subsequent choices.

From the perspective of employees, the choice of a more comprehensive solution has higher performance, but at the same time the complexity of the functions will be relatively high, requiring the company team to spend more time learning and adapting. Even due to the complexity of the functions, it is often necessary to hire external consultants, which will bring additional costs. The 'good enough' solution, although it only has simple functions, can make the employee team get started more quickly at the beginning, reduce the cost of employee training and learning, and allow employees to gradually learn when the subsequent solution is upgraded, reducing pressure.

Finally, we consider virtualization technology and performance from a comprehensive solution perspective. Many advanced features such as DRS and DR have been improved, but they are not necessary for small businesses. These are all additional costs, and a 'good enough' solution that provides basic functions such as virtual machine construction and operation can already meet most virtualization needs.

Overall, as a small business just starting out, a 'good enough' solution is more suitable. The high cost of a comprehensive solution will bring more hidden dangers. At the same time, there will be a lot of redundant functions that will not be used in the short term. We can choose a 'good enough' solution and avoid risks by selectively upgrading and expanding the solution as the company expands.

Reference:

1. Admin. (2023). Unlock the power of network virtualization: Exploring application scenarios. Retrieved from <https://ciohub.org/post/2023/09/unlocking-network-virtualization-application-scenarios/>
2. Alibaba Cloud. (2024). Emulating hardware: CPU virtualization under the hood. Retrieved from <https://www.alibabacloud.com/tech-news/a/cpu-virtualization/4okq5h7cc14-emulating-hardware-cpu-virtualization-under-the-hood#:~:text=This%20allows%20software%20written%20for,need%20for%20the%20original%20hardware.>
3. AWS. (n.d.). The difference between Type 1 and Type 2 hypervisors. Retrieved from <https://aws.amazon.com/compare/the-difference-between-type-1-and-type-2-hypervisors/#:~:text=A%20type%20%20hypervisor%2C%20or,to%20obtain%20underlying%20system%20resources.>
4. GeeksforGeeks. (2023). Moore's law. Retrieved from <https://www.geeksforgeeks.org/moores-law/>
5. Hashemi-Pour, C., Brush, K., & Kirsch, B. (2024). Virtualization. Retrieved from <https://www.techtarget.com/searchitoperations/definition/virtualization>
6. Susnjara, S. & Smalley, I. (2024). What are Hypervisors? Retrieved from <https://www.ibm.com/think/topics/hypervisors>
7. Ribeiro, A. (2024). CPR data reports 32% rise this year as global healthcare sector faces surge in cyberattacks. Retrieved from <https://industrialcyber.co/medical/cpr-data-reports-32-rise-this-year-as-global-healthcare-sector-faces-surge-in-cyberattacks/>
8. JBS.live. (n.d.). Explaining the role of the hypervisor in virtualization. Retrieved from <https://jbs.live/explaining-the-role-of-the-hypervisor-in-virtualization/#:~:text=Emulation%20of%20Hardware:%20The%20hypervisor,a%20failure%20or%20data%20loss.>
9. LinkedIn. (n.d.). How do you solve software compatibility? Retrieved from <https://www.linkedin.com/advice/1/how-do-you-solve-software-compatibility#:~:text=Compatibility%20issues%20occur%20when%20software,performance%20problems%2C%20or%20security%20risks.>

10. LinkedIn. (n.d.). How does a hypervisor improve security isolation? Retrieved from <https://www.linkedin.com/advice/0/how-does-hypervisor-improve-security-isolation#:~:text=The%20hypervisor%20can%20isolate%20the,the%20VMs%20before%20they%20start.&text=From%20a%20hardware%20perspective%20CPU,a%20hardware%20root%20of%20trust.>
11. LinkedIn. (n.d.). What are the pros and cons of isolating vs eradicating infected devices? Retrieved from <https://www.linkedin.com/advice/0/what-pros-cons-isolating-vs-eradicating-infected>
12. Patel, R. (2024). Understanding hypervisors: Exploring Type 1 vs Type 2 and full vs para-virtualization. Retrieved from <https://medium.com/@ravipatel.it/understanding-hypervisors-exploring-type-1-vs-type-2-and-full-vs-para-virtualization-71b4dad9abd9>
13. Network Urge. (2017). What is a hypervisor? Retrieved from <https://www.networkurge.com/2017/06/what-is-hypervisor.html>
14. SentinelOne. (2024). What is a Hypervisor? Types, Benefits, and Best Practices. Retrieved from <https://www.sentinelone.com/cybersecurity-101/cybersecurity/what-is-a-hypervisor/#:~:text=There%20are%20two%20types%20of,scalability%2C%20or%20ease%20of%20use.>
15. Technologies, S. (2024). Type 1 vs. Type 2 hypervisor: Key differences. Retrieved from <https://www.sangfor.com/glossary/cloud-and-infrastructure/type-1-vs-type-2-hypervisor-key-differences#:~:text=Enterprise%20workloads%20requiring%20maximum%20performance,demand%20consistent%20and%20reliable%20performance.>
16. Pankevych, O. (2023). Type 1 vs. Type 2 hypervisor: What is the difference? Retrieved from <https://www.starwindsoftware.com/blog/type-1-vs-type-2-hypervisor-what-is-the-difference/>
17. Bigelow, S. & Kirsch, B. (2024). What's the difference between Type 1 vs Type 2 hypervisor? Retrieved from <https://www.techtarget.com/searchitoperations/tip/Whats-the-difference-between-Type-1-vs-Type-2-hypervisor>

18. Terranova Security. (2024). The most dangerous healthcare cyber attacks. Retrieved from <https://www.terrnovasecurity.com/blog/most-dangerous-healthcare-cyber-attacks>
19. Wikipedia contributors. (2024). Virtualization. Retrieved from <https://en.wikipedia.org/wiki/Virtualization>
20. Wind River. (n.d.). Hypervisor. Retrieved from <https://www.windriver.com/solutions/learning/hypervisor>

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